

Database Management Systems

Course Metadata

Credit Structure	L-T-P: 4-0-0 Total: 4
Offering Semester	4 th Semester, Spring
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	NIL
Faculty Names	Plaban Kumar Bhowmick, Jiaul H Paik, Koustav Rudra
Overlapping Subjects	DBMS CS <5%
Justification of overlap	This course is a core for the B Tech in AI program

Course Description

Data is an integral ingredient of the artificial intelligence systems and models. Development of practical and real-world AI systems hence calls for efficient modelling, storage and retrieval of data. This course explores different avenues of database management. Particularly, three database design paradigms have been discussed, namely, Relational database, NoSQL and Graph database design.

Course Content

- Introduction to database systems
- Relational Database Management Systems: Data modelling techniques, querying relational data, query evaluation, relational data storage and transaction management.
- NoSQL Database: NoSQL database design philosophy and building blocks, NoSQL database design with MongoDB
- Graph Databases: Two types of graph databases, namely, RDF store and Labelled property graphs, querying graph databases with SPARQL and Cypher

Sl No.	Module	Unit	Module Hr (48)
Part-1: Relational Database (28 hrs)			
1	Introduction to RDBMS and Data Model	Overview of DBMS, Entity-Relationship, Relationships, Integrity constraints, Unified Modelling Language (UML)	3
2	Querying Relational Data	Relational algebra and operations, tuple and domain calculus Structured Query Language (SQL): basic SQL query, aggregate operators, Null values, Integrity constraints, triggers	6
3	Relational Database Design	Domain and data dependency, Schema refinement and normal forms	4
4	Storing and Indexing Data	Tree structured indexing (B+ Trees) Hash-based indexing (static hashing, extendible hashing, linear hashing)	4
5	Query Evaluation	Evaluating relational operators and expressions, Translating SQL queries into algebra, relational algebra equivalences Relational query optimizer	6
6	Transaction Management	The ACID properties, Transactions and schedules, concurrent execution, lock-based concurrency control, transaction support in SQL, crash recovery	5
Part-2: NoSQL Database (10 hrs)			
7	Introduction to NoSQL	Motivation for NoSQL databases, CAP theorem, ACID vs BASE, Types of NoSQL databases (Key-Value pair, document, column family store, graph databases)	2
8	Basics of Document Database	Document as JSON, Operations on Document Databases, Collections, Embedded document, Schemaless design, Polymorphic schema, Partitions, Sharding, Data distribution algorithm, Data modelling and query processing, document database design	4
9	MongoDB	Basic operations, management and querying, Collection and data aggregation, fault tolerance and scalability (replica management, sharding), database administration	4
Part-3: Graph Database (10 hrs)			
10	RDF Store	Triple data model, SPARQL query language, SPARQL algebra, storage and indexing	5

11	Labelled Property Model	Graph data model, Labelled property graph, elements of graph data application, graph data storage and indexing, Neo4j graph database, querying graph database with Cypher, Predictive analysis on graph data	5
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Books

1. **Database System Concepts, Deep Learning on Graphs**, Abraham Silberschatz, Henry Korth, and S. Sudarshan, McGrawHill, 6th Edition, 2013
2. **Fundamentals of Database Systems**, R. Elmasri and S. Navathe, 7th Edition, Pearson, 2017
3. **Database Management Systems**, Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition, McGraw-Hill, 2002
4. **MongoDB: The Definitive Guide**, Shannon Bradshaw, Eoin Brazil & Kristina Chodorow, 3rd Edition, O'Reilly, 2019
5. **Graph Databases**, Ian Robinson, Jim Webber & Emil Eifrem, 2nd Edition, O'Reilly, 2015
6. **RDF Database Systems**, Olivier Cure and Guillaume Blin, 1st Edition, Morgan and Kaufmann, 2015